

The Course

Synthetic Aperture Radar (SAR)

Part 7: SAR Polarimetry 2

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

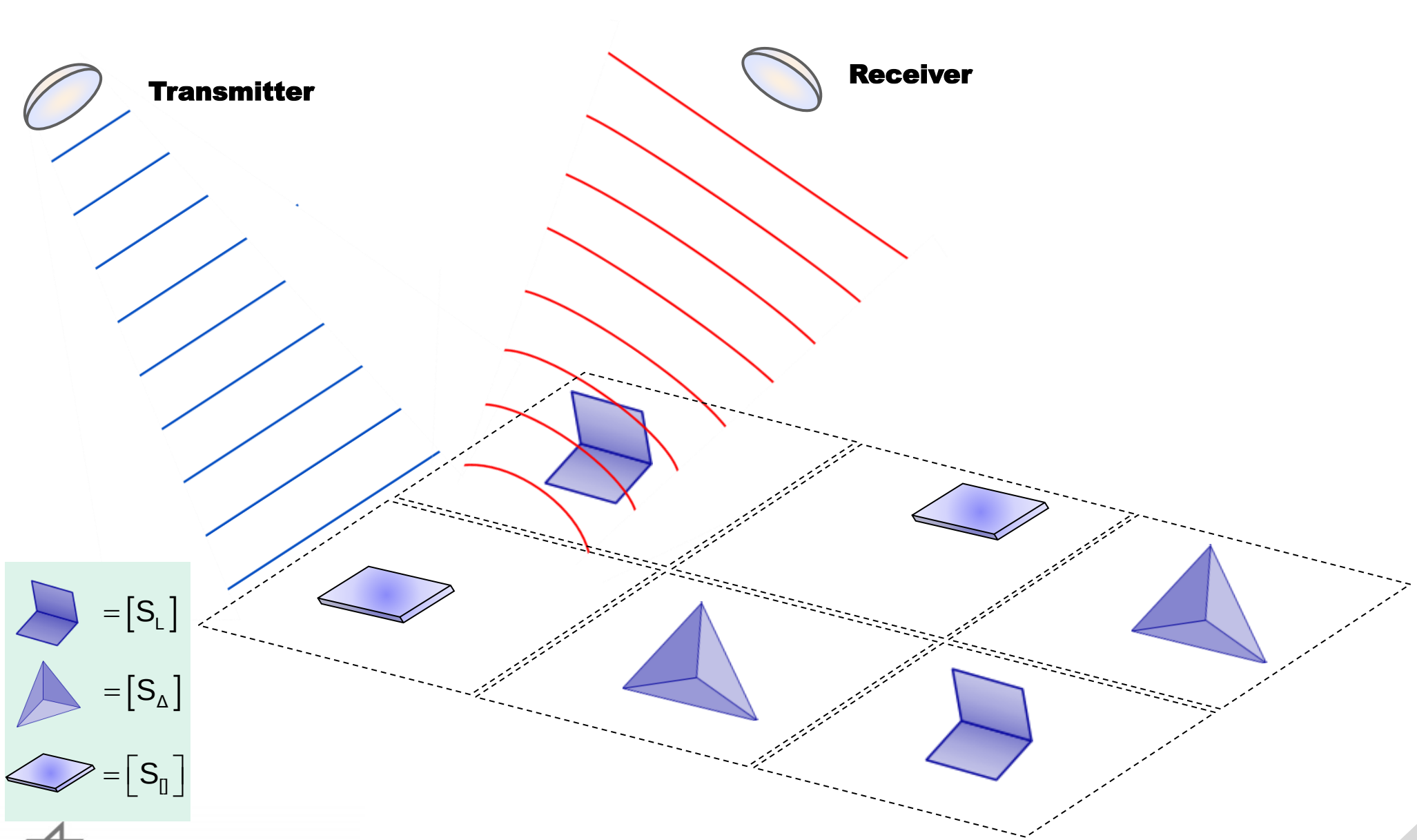
Email: noelia.romerpuig@dlr.de, paloma.saporta@dlr.de, kostas.papathanassiou@dlr.de



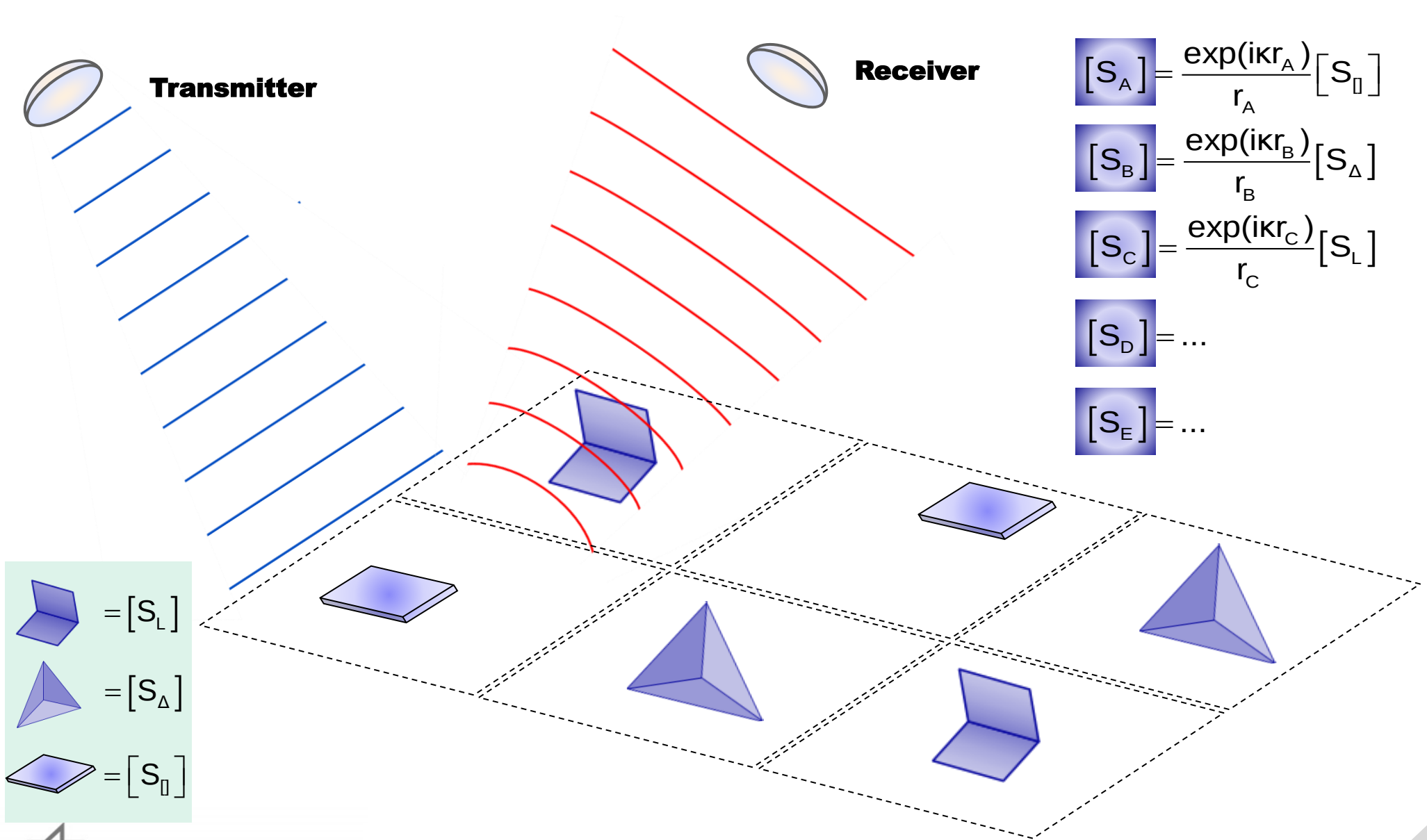
ESAMAAP

EEBIOMASS

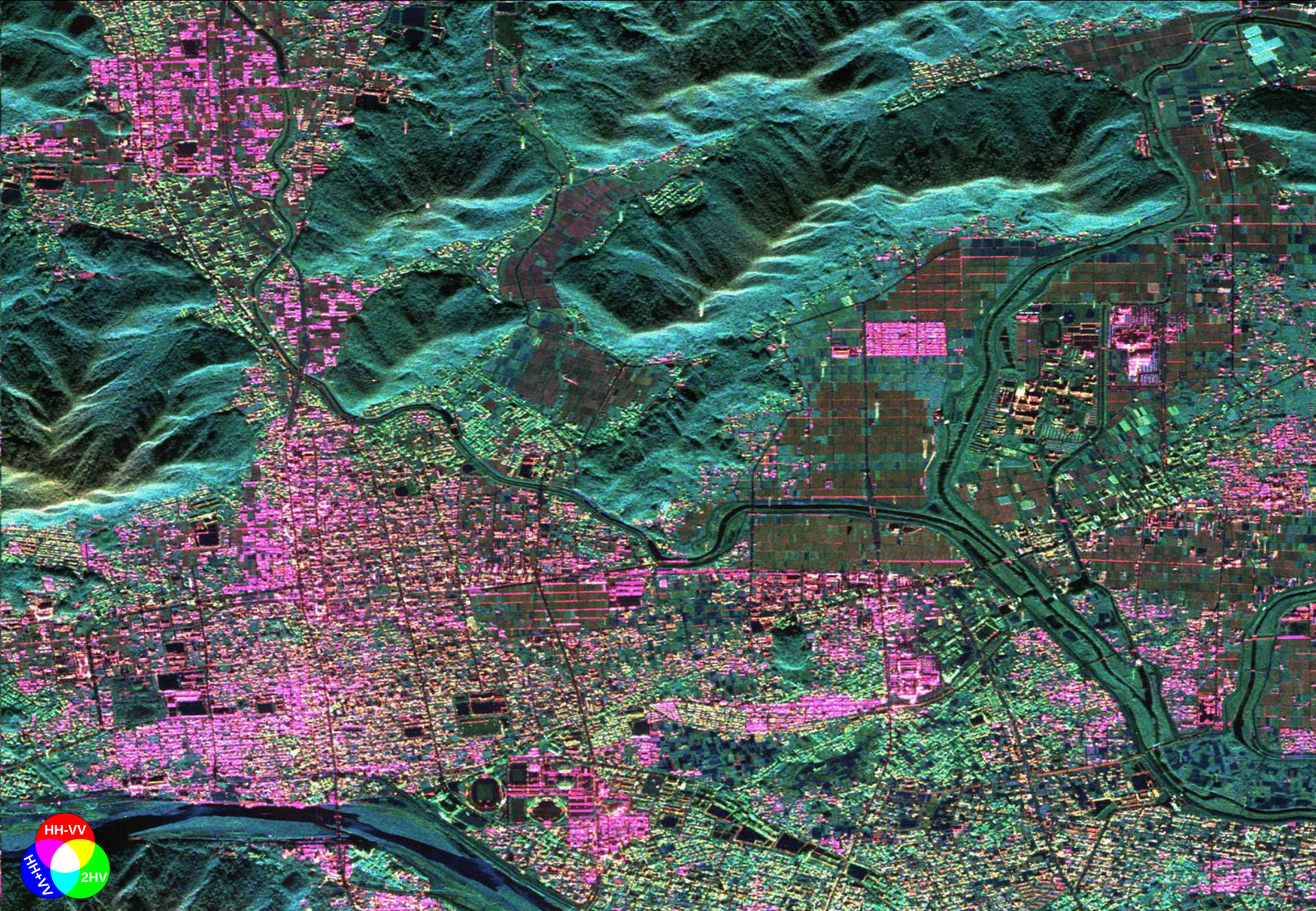
Deterministic Scatterers: Non-Depolarising Scatterers



Deterministic Scatterers: Non-Depolarising Scatterers

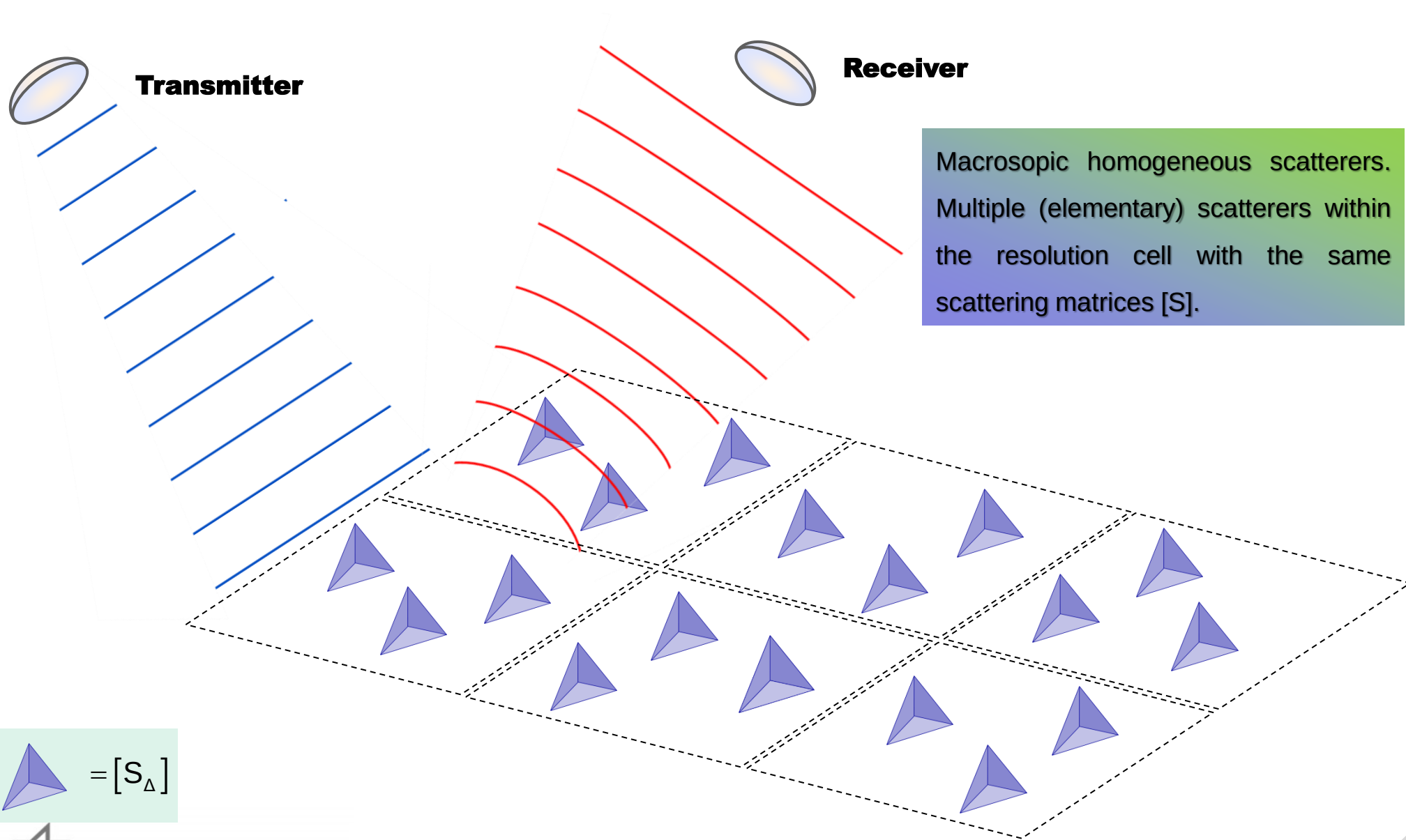




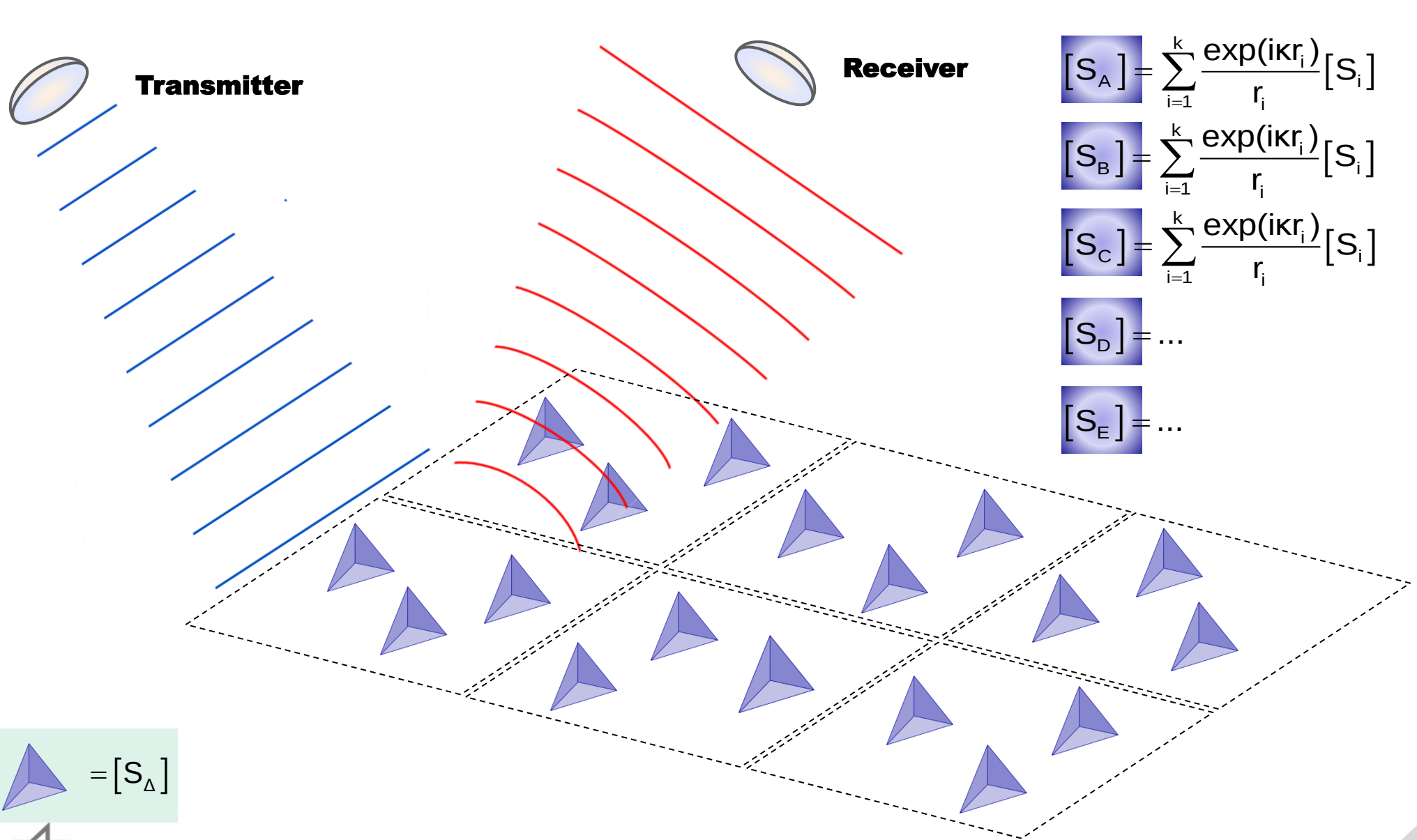




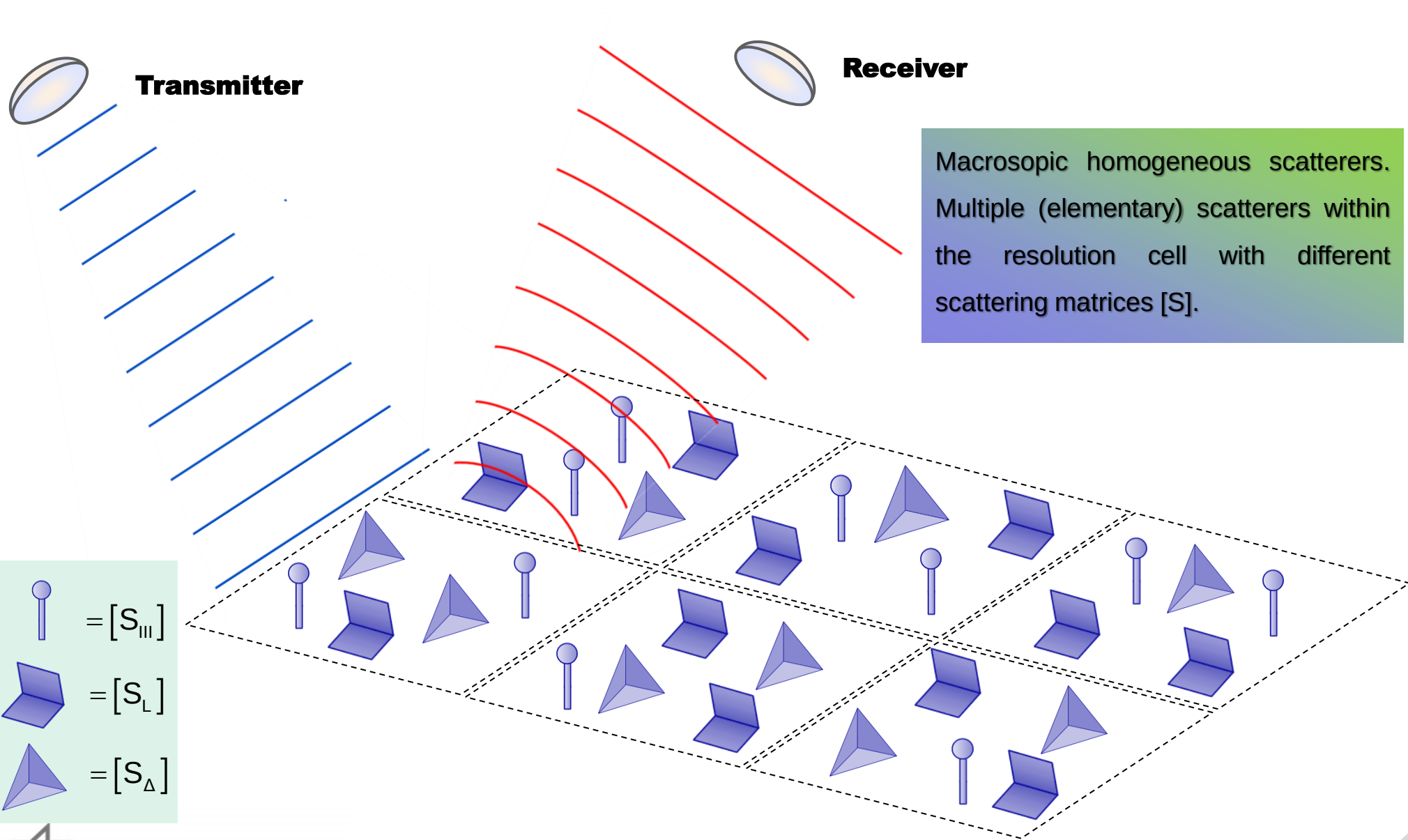
Distributed Scatterers - Case 1: Non-Depolarising Scatterers



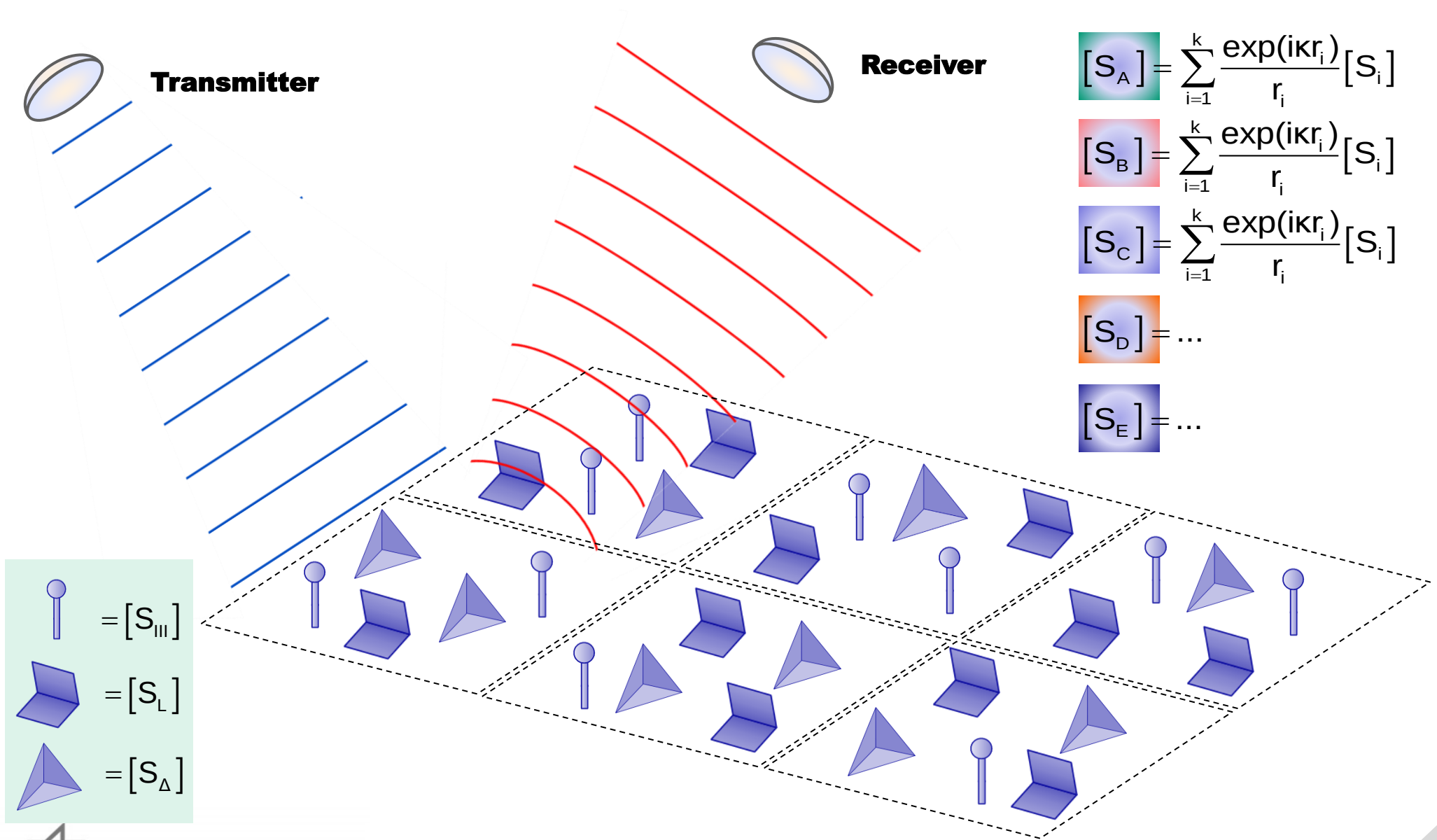
Distributed Scatterers - Case 1: Non-Depolarising Scatterers



Distributed Scatterers - Case 2: Depolarising Scatterers



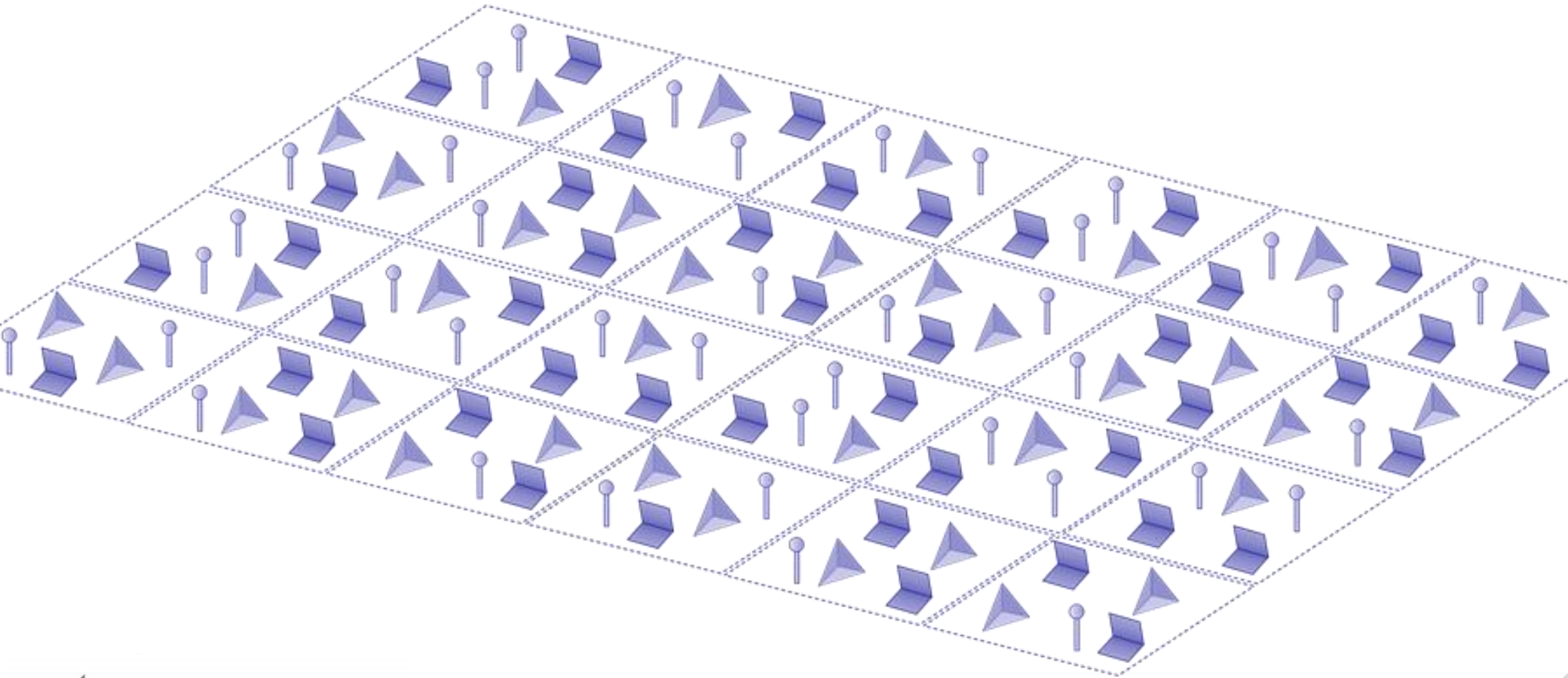
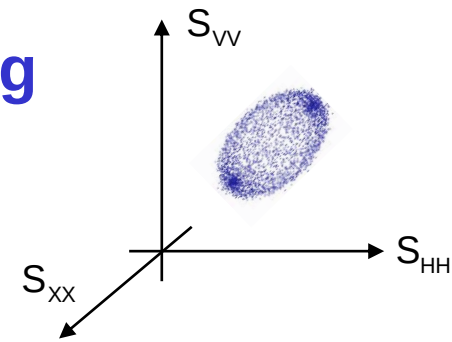
Distributed Scatterers - Case 2: Depolarising Scatterers



Covariance & Coherency Matrices in Backscattering

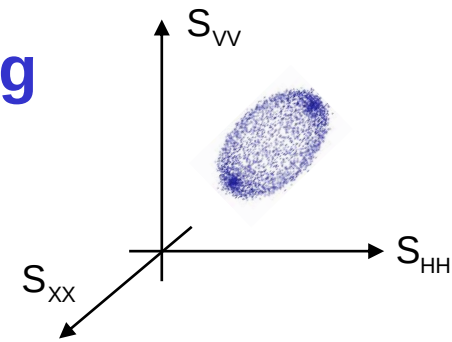
$$[S_i] = \begin{bmatrix} S_{HH_i} & S_{HV_i} \\ S_{VH_i} & S_{VV_i} \end{bmatrix} \quad \rightarrow \quad \vec{k}_{3L_i} = \begin{bmatrix} S_{HH_i} \\ \sqrt{2} \cdot S_{XX_i} \\ S_{VV_i} \end{bmatrix}$$

Lexicographic Scattering Vector



Covariance & Coherency Matrices in Backscattering

$$[S_i] = \begin{bmatrix} S_{HH_i} & S_{HV_i} \\ S_{VH_i} & S_{VV_i} \end{bmatrix} \rightarrow \vec{k}_{3L_i} = \begin{bmatrix} S_{HH_i} \\ \sqrt{2} \cdot S_{XX_i} \\ S_{VV_i} \end{bmatrix} \quad \text{Lexicographic Scattering Vector}$$



Covariance Matrix:

$$[C_3] := \langle \vec{k}_{3L_i} \cdot \vec{k}_{3L_i}^* \rangle = \begin{bmatrix} \langle |S_{HH_i}|^2 \rangle & \sqrt{2} \langle S_{HH_i} S_{HV_i}^* \rangle & \langle S_{HH_i} S_{VV_i}^* \rangle \\ \sqrt{2} \langle S_{HV_i} S_{HH_i}^* \rangle & 2 \langle |S_{HV_i}|^2 \rangle & \sqrt{2} \langle S_{HV_i} S_{VV_i}^* \rangle \\ \langle S_{VV_i} S_{HH_i}^* \rangle & \sqrt{2} \langle S_{VV_i} S_{HV_i}^* \rangle & \langle |S_{VV_i}|^2 \rangle \end{bmatrix}$$

$$[S_i] = \begin{bmatrix} S_{HH_i} & S_{HV_i} \\ S_{VH_i} & S_{VV_i} \end{bmatrix} \rightarrow \vec{k}_{3P_i} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH_i} + S_{VV_i} \\ S_{HH_i} - S_{VV_i} \\ 2S_{XX_i} \end{bmatrix} \quad \text{Pauli Scattering Vector}$$

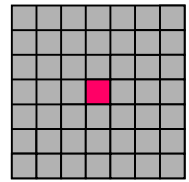
Coherency Matrix:

$$[T_3] := \langle \vec{k}_{3P_i} \cdot \vec{k}_{3P_i}^* \rangle = \begin{bmatrix} \langle |S_{HH_i} + S_{VV_i}|^2 \rangle & \langle (S_{HH_i} + S_{VV_i})(S_{HH_i} - S_{VV_i})^* \rangle & 2 \langle (S_{HH_i} + S_{VV_i}) S_{HV_i}^* \rangle \\ \langle (S_{HH_i} - S_{VV_i})(S_{HH_i} + S_{VV_i})^* \rangle & \langle |S_{HH_i} - S_{VV_i}|^2 \rangle & 2 \langle (S_{HH_i} - S_{VV_i}) S_{HV_i}^* \rangle \\ 2 \langle S_{HV_i} (S_{HH_i} + S_{VV_i})^* \rangle & 2 \langle S_{HV_i} (S_{HH_i} - S_{VV_i})^* \rangle & 4 \langle |S_{HV_i}|^2 \rangle \end{bmatrix}$$

$[C_3]$ and $[T_3]$ are by definition hermitian positive semi-definite matrices (i.e. have positive eigenvalues).

Both contain in general **9 independent parameters: 3 real diagonal elements (powers) and 3 complex off-diagonal elements**. They describe completely depolarising scatters.

Eigen-Decomposition: Eigenvalues and Eigenvectors



Scattering Matrix $[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \Rightarrow \vec{k}_{3P} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{XX} \end{bmatrix} \Rightarrow [T_3] := \langle \vec{k}_{3P} \cdot \vec{k}_{3P}^+ \rangle$ Coherence Matrix

The Eigen-Decomposition (PCA) allows to express the coherence (covariance) matrix in terms of up to three uncorrelated (e.g. orthogonal) scattering contributions:

Eigen Decomposition:
(PCA or Diagonalisation)

$$[T_3] = [U_3] [\Lambda] [U_3]^{-1} \longrightarrow \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^* & T_{22} & T_{23} \\ T_{13}^* & T_{23}^* & T_{33} \end{bmatrix} = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} e_{11}^* & e_{21}^* & e_{31}^* \\ e_{12}^* & e_{22}^* & e_{32}^* \\ e_{13}^* & e_{23}^* & e_{33}^* \end{bmatrix}$$

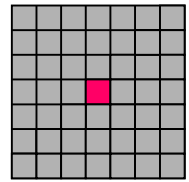
($[\Lambda] = [U_3]^{-1} [T_3] [U_3]$)

► 3 orthonormal Eigenvectors $\vec{e}_1 = \begin{bmatrix} e_{11} \\ e_{21} \\ e_{31} \end{bmatrix}$ $\vec{e}_2 = \begin{bmatrix} e_{12} \\ e_{22} \\ e_{32} \end{bmatrix}$ $\vec{e}_3 = \begin{bmatrix} e_{13} \\ e_{23} \\ e_{33} \end{bmatrix}$ represent the scattering contributions

► 3 real positive Eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ express the “importance” of the scattering contributions

$$[T_3] = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^* & T_{22} & T_{23} \\ T_{13}^* & T_{23}^* & T_{33} \end{bmatrix} \quad [\Lambda] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad [U_3] = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \quad [U_3]^{-1} = [U_3]^+ = \begin{bmatrix} e_{11}^* & e_{21}^* & e_{31}^* \\ e_{12}^* & e_{22}^* & e_{32}^* \\ e_{13}^* & e_{23}^* & e_{33}^* \end{bmatrix}$$

Eigen-Decomposition: Eigenvalues and Eigenvectors



Scattering Matrix $[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \Rightarrow \vec{k}_{3P} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{XX} \end{bmatrix} \Rightarrow [T_3] := \langle \vec{k}_{3P} \cdot \vec{k}_{3P}^+ \rangle$ Coherence Matrix

Eigen
Decomposition:
(PCA or Diagonalisation)

$$[T_3] = [U_3] [\Lambda] [U_3]^{-1} \longrightarrow \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^* & T_{22} & T_{23} \\ T_{13}^* & T_{23}^* & T_{33} \end{bmatrix} = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} e_{11}^* & e_{21}^* & e_{31}^* \\ e_{12}^* & e_{22}^* & e_{32}^* \\ e_{13}^* & e_{23}^* & e_{33}^* \end{bmatrix}$$

$$\begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^* & T_{22} & T_{23} \\ T_{13}^* & T_{23}^* & T_{33} \end{bmatrix} = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \left(\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \right) \begin{bmatrix} e_{11}^* & e_{21}^* & e_{31}^* \\ e_{12}^* & e_{22}^* & e_{32}^* \\ e_{13}^* & e_{23}^* & e_{33}^* \end{bmatrix}$$

Eigen
Decomposition:

$$[T_3] = [U_3] [\Lambda] [U_3]^{-1} = \sum_{i=1}^3 \lambda_i (\vec{e}_i \cdot \vec{e}_i^+) = \lambda_1 (\vec{e}_1 \cdot \vec{e}_1^+) + \lambda_2 (\vec{e}_2 \cdot \vec{e}_2^+) + \lambda_3 (\vec{e}_3 \cdot \vec{e}_3^+) = \lambda_1 [T_3^1] + \lambda_2 [T_3^2] + \lambda_3 [T_3^3]$$

► 3 orthonormal Eigenvectors $\vec{e}_1 = \begin{bmatrix} e_{11} \\ e_{21} \\ e_{31} \end{bmatrix}$ $\vec{e}_2 = \begin{bmatrix} e_{12} \\ e_{22} \\ e_{32} \end{bmatrix}$ $\vec{e}_3 = \begin{bmatrix} e_{13} \\ e_{23} \\ e_{33} \end{bmatrix}$ represent the scattering contributions

► 3 real positive Eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ express the “importance” of the scattering contributions

Eigen-Decompositions: Scattering Entropy and Anisotropy

$$[T_3] = [U_3][\Lambda][U_3]^{-1} = \sum_{i=1}^3 \lambda_i (\vec{e}_i \cdot \vec{e}_i^+) = \lambda_1 (\vec{e}_1 \cdot \vec{e}_1^+) + \lambda_2 (\vec{e}_2 \cdot \vec{e}_2^+) + \lambda_3 (\vec{e}_3 \cdot \vec{e}_3^+) = \lambda_1 [T_3^1] + \lambda_2 [T_3^2] + \lambda_3 [T_3^3]$$

Scattering Entropy: $H := \sum_{i=1}^3 P_i \log_3 P_i$ with $P_i := \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$

general measure of scattering depolarisation.

- $0 \leq H \leq 1$ where:
- $H = 0 \rightarrow$ Totally Polarised Scatterer : $\lambda_1 = TP$ and $\lambda_2 = \lambda_3 = 0$
 - $H = 1 \rightarrow$ Totally Unpolarised Scatterer : $\lambda_1 = \lambda_2 = \lambda_3$

Scattering Anisotropy: $A := \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3} = \frac{P_2 - P_3}{P_2 + P_3}$ with $P_i := \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$

provides information about the variation of depolarisation with changing polarization.

- $0 \leq A \leq 1$ where:
- $A = 0 \rightarrow$ 2 Equal Secondary Scattering Processes: $\lambda_2 = \lambda_3$
 - $A = 1 \rightarrow$ Only 1 Secondary Scattering Process:

$A=0$: depolarisation remains the same across all polarimetric sub-spaces;

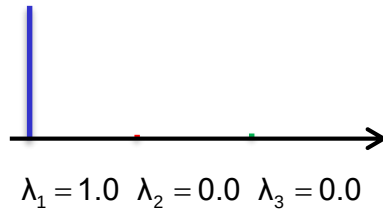
$A=1$: there are polarimetric subspaces with low(er) depolarisation.

Eigen-Decompositions: Scattering Entropy and Anisotropy

$$[T_3] = [U_3][\Lambda][U_3]^{-1} = \sum_{i=1}^3 \lambda_i(\vec{e}_i \cdot \vec{e}_i^+) = \lambda_1(\vec{e}_1 \cdot \vec{e}_1^+) + \lambda_2(\vec{e}_2 \cdot \vec{e}_2^+) + \lambda_3(\vec{e}_3 \cdot \vec{e}_3^+) = \lambda_1[T_3^1] + \lambda_2[T_3^2] + \lambda_3[T_3^3]$$

Eigenvalues derived parameters

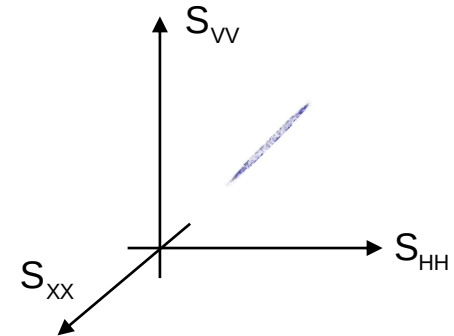
Single Dominant Scatterer:



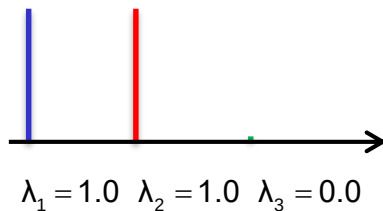
$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = 0$$

$$P_1 = 1, \quad P_2 = 0, \quad P_3 = 0$$

$$H = 0, \quad A = 0$$



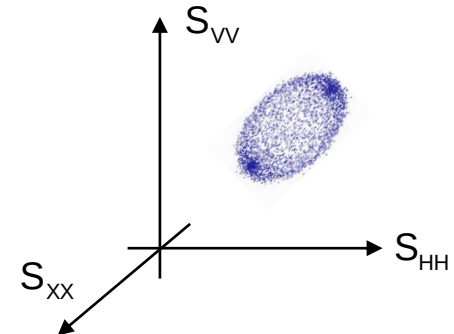
Two Significant Scatterers:



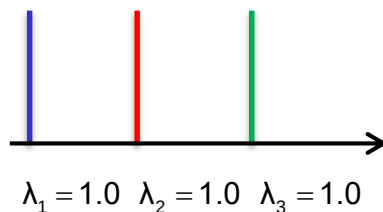
$$\lambda_1 = 1, \quad \lambda_2 = 1, \quad \lambda_3 = 0$$

$$P_1 = .5, \quad P_2 = .5, \quad P_3 = 0$$

$$H = .65, \quad A = 1.$$



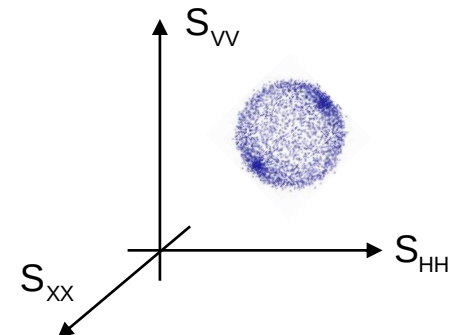
Three (or more) Significant Scatterers:



$$\lambda_1 = 1, \quad \lambda_2 = 1, \quad \lambda_3 = 1$$

$$P_1 = .33, \quad P_2 = .33, \quad P_3 = .33$$

$$H = 1., \quad A = 0.$$

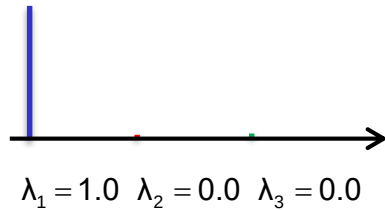


Eigen-Decompositions: Scattering Entropy and Anisotropy

$$[T_3] = [U_3][\Lambda][U_3]^{-1} = \sum_{i=1}^3 \lambda_i(\vec{e}_i \cdot \vec{e}_i^+) = \lambda_1(\vec{e}_1 \cdot \vec{e}_1^+) + \lambda_2(\vec{e}_2 \cdot \vec{e}_2^+) + \lambda_3(\vec{e}_3 \cdot \vec{e}_3^+) = \lambda_1[T_3^1] + \lambda_2[T_3^2] + \lambda_3[T_3^3]$$

Eigenvalues derived parameters

Single Dominant Scatterer:

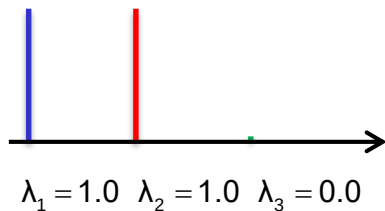


$$\lambda_1 = 1, \lambda_2 = 0, \lambda_3 = 0$$

$$P_1 = 1, P_2 = 0, P_3 = 0$$

$$H = 0, A = 0$$

Two Significant Scatterers:

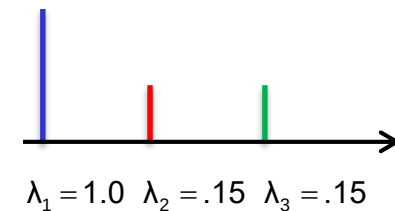


$$\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = 0$$

$$P_1 = .5, P_2 = .5, P_3 = 0$$

$$H = .65, A = 1.$$

Single Significant Scatterer + Noise:

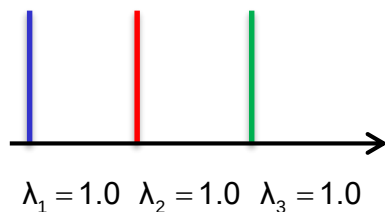


$$\lambda_1 = 1, \lambda_2 = .15, \lambda_3 = .15$$

$$P_1 = .76, P_2 = .12, P_3 = .12$$

$$H = .65, A = 0.$$

Three (or more) Significant Scatterers:



$$\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = 1$$

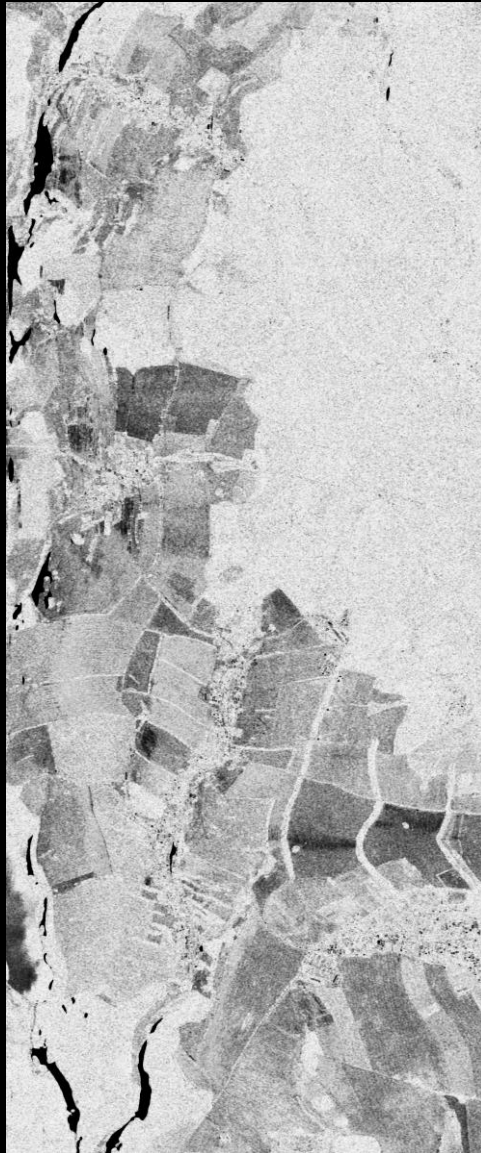
$$P_1 = .33, P_2 = .33, P_3 = .33$$

$$H = 1., A = 0.$$

Eigen Decomposition



Pauli RGB



Entropy H

H=0

H=1



Anisotropy A

A=0

A=1

Eigen-Decompositions: Mean Scattering Mechanism

$$[T_3] = [U_3][\Lambda][U_3]^{-1} = \sum_{i=1}^3 \lambda_i (\vec{e}_i \cdot \vec{e}_i^+) = \lambda_1 (\vec{e}_1 \cdot \vec{e}_1^+) + \lambda_2 (\vec{e}_2 \cdot \vec{e}_2^+) + \lambda_3 (\vec{e}_3 \cdot \vec{e}_3^+) = \lambda_1 [T_3^1] + \lambda_2 [T_3^2] + \lambda_3 [T_3^3]$$

$$[U_3] = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} | & | & | \\ \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ | & | & | \end{bmatrix} \quad \text{Unitary Representation: } \vec{e} = \begin{bmatrix} \cos(\alpha) \exp(i\gamma) \\ \sin(\alpha) \cos(\beta) \exp(i\delta) \\ \sin(\alpha) \sin(\beta) \exp(i\varepsilon) \end{bmatrix}$$



$$[U_3] = \begin{bmatrix} \cos(\alpha_1) \exp(i\gamma_1) & \cos(\alpha_2) \exp(i\gamma_2) & \cos(\alpha_3) \exp(i\gamma_3) \\ \sin(\alpha_1) \cos(\beta_1) \exp(i\delta_1) & \sin(\alpha_2) \cos(\beta_2) \exp(i\delta_2) & \sin(\alpha_3) \cos(\beta_3) \exp(i\delta_3) \\ \sin(\alpha_1) \sin(\beta_1) \exp(i\varepsilon_1) & \sin(\alpha_2) \sin(\beta_2) \exp(i\varepsilon_2) & \sin(\alpha_3) \sin(\beta_3) \exp(i\varepsilon_3) \end{bmatrix}$$

**1st Scattering
Mechanism ($[T_1]$)**

**2nd Scattering
Mechanism ($[T_2]$)**

**3rd Scattering
Mechanism ($[T_3]$)**

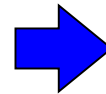
Mean α -Angle: $\alpha = P_1 \alpha_1 + P_2 \alpha_2 + P_3 \alpha_3$

Mean β -Angle: $\beta = P_1 \beta_1 + P_2 \beta_2 + P_3 \beta_3$

Mean γ -Angle: $\gamma = P_1 \gamma_1 + P_2 \gamma_2 + P_3 \gamma_3$

Mean δ -Angle: $\delta = P_1 \delta_1 + P_2 \delta_2 + P_3 \delta_3$

Mean ε -Angle: $\varepsilon = P_1 \varepsilon_1 + P_2 \varepsilon_2 + P_3 \varepsilon_3$



$$\vec{e} = \begin{bmatrix} \cos(\alpha) \exp(i\gamma) \\ \sin(\alpha) \cos(\beta) \exp(i\delta) \\ \sin(\alpha) \sin(\beta) \exp(i\varepsilon) \end{bmatrix}$$

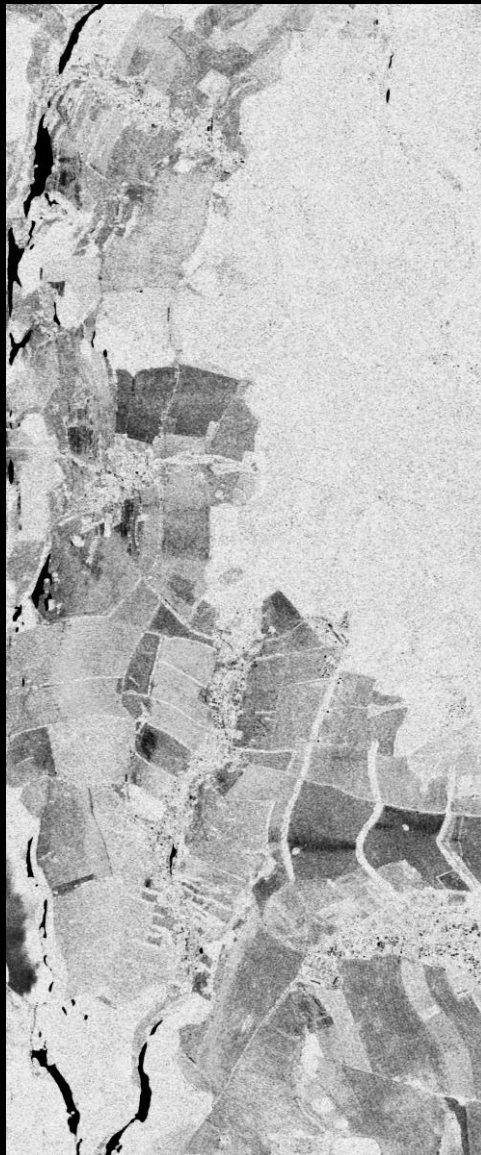
with $P_i := \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$

Mean Scattering Mechanism

Eigen Decomposition



Pauli RGB



Entropy H

H=0

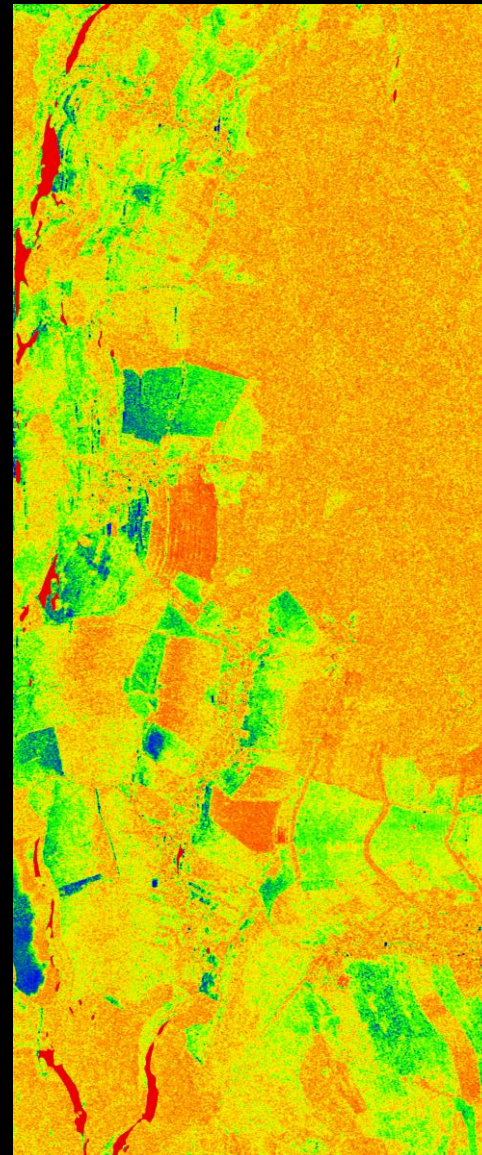
H=1



Anisotropy A

A=0

A=1

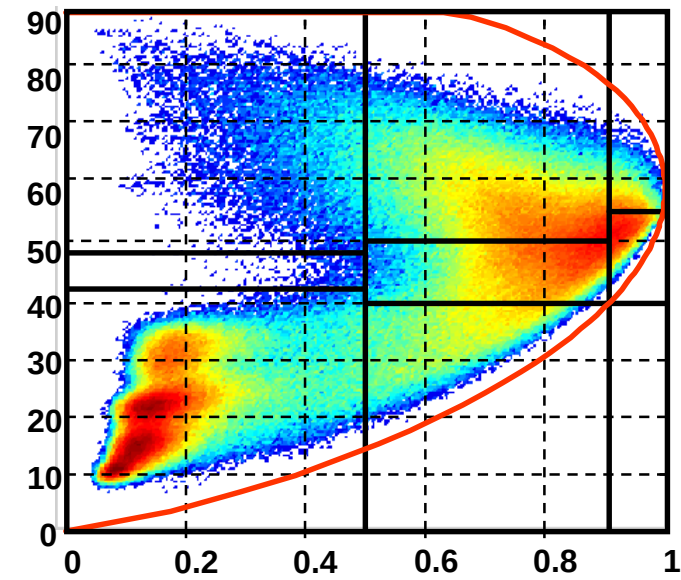
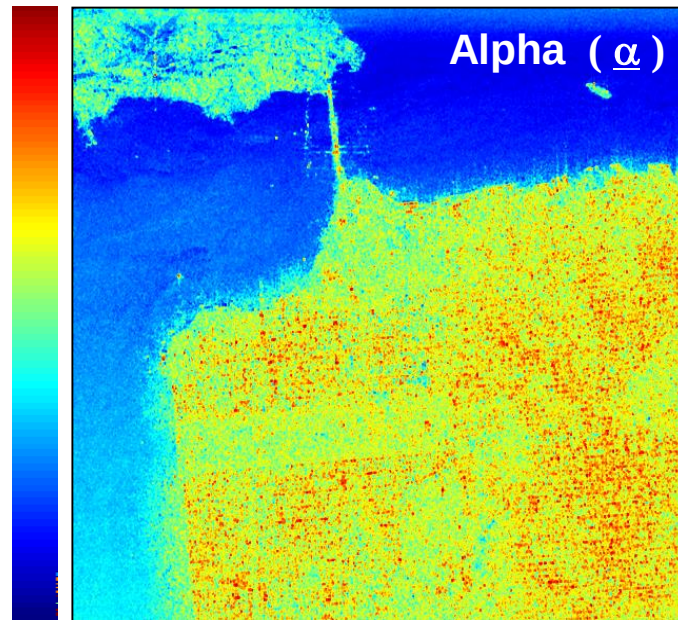
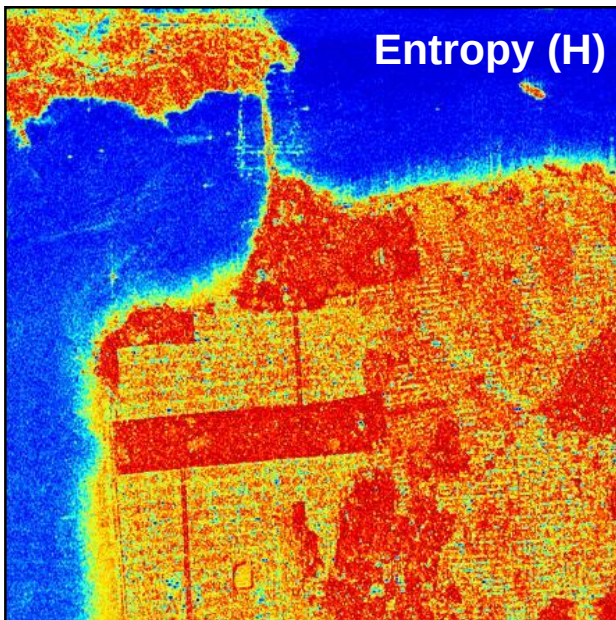
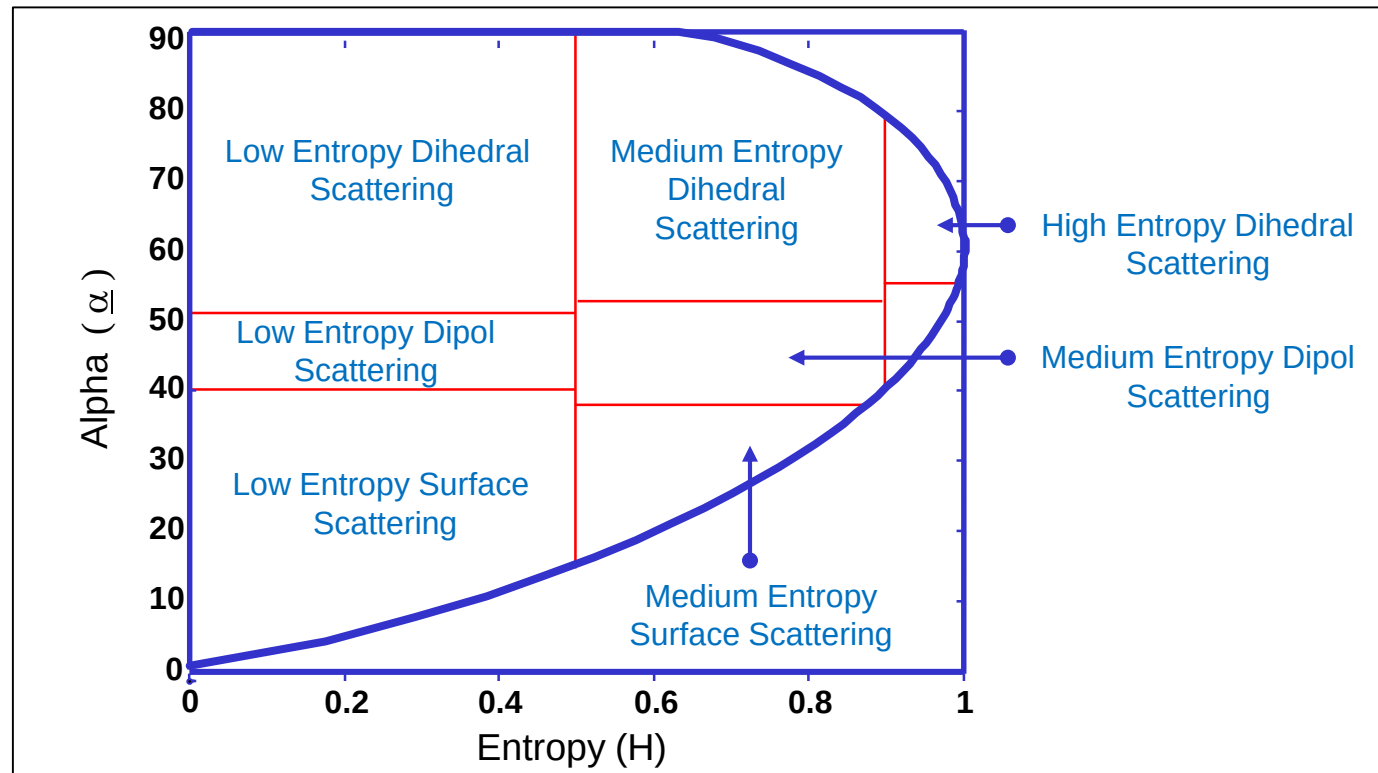
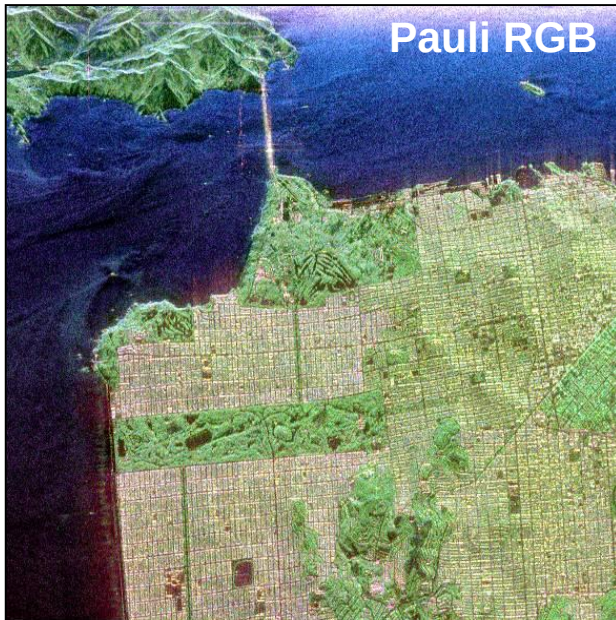


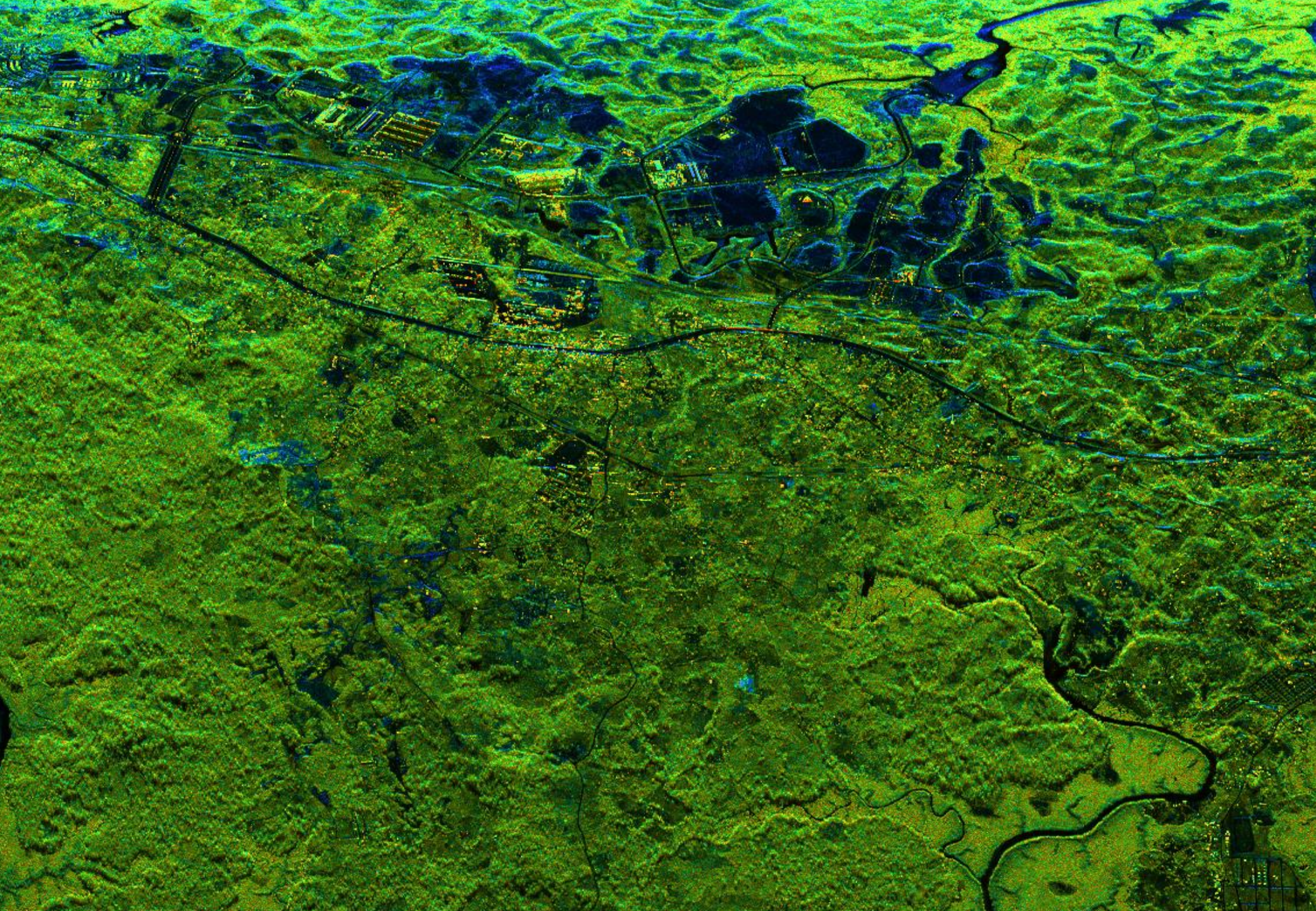
Alpha Angle α (Mean)

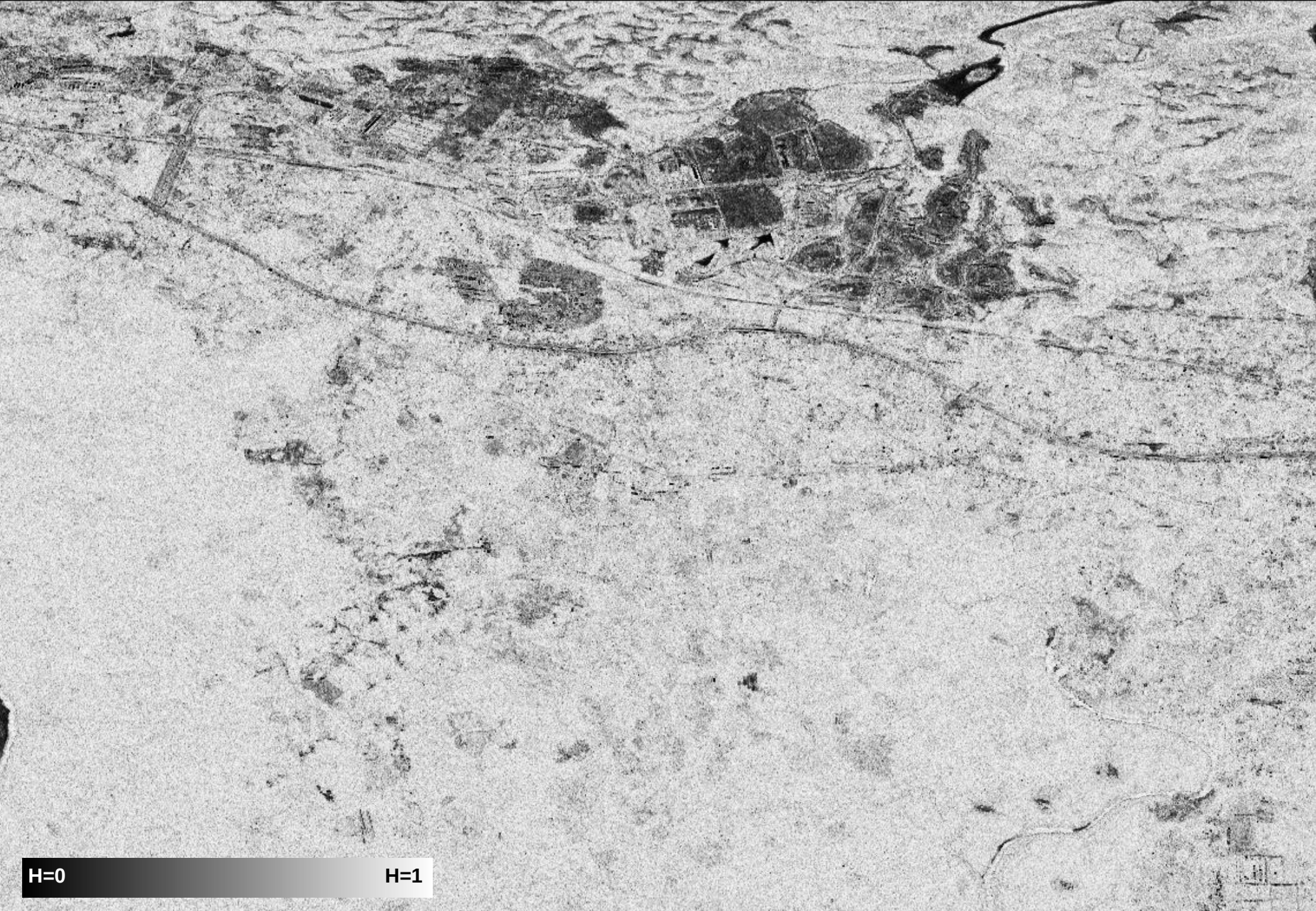
$\alpha=0^\circ$

$\alpha=90^\circ$

H - α Angle Plane







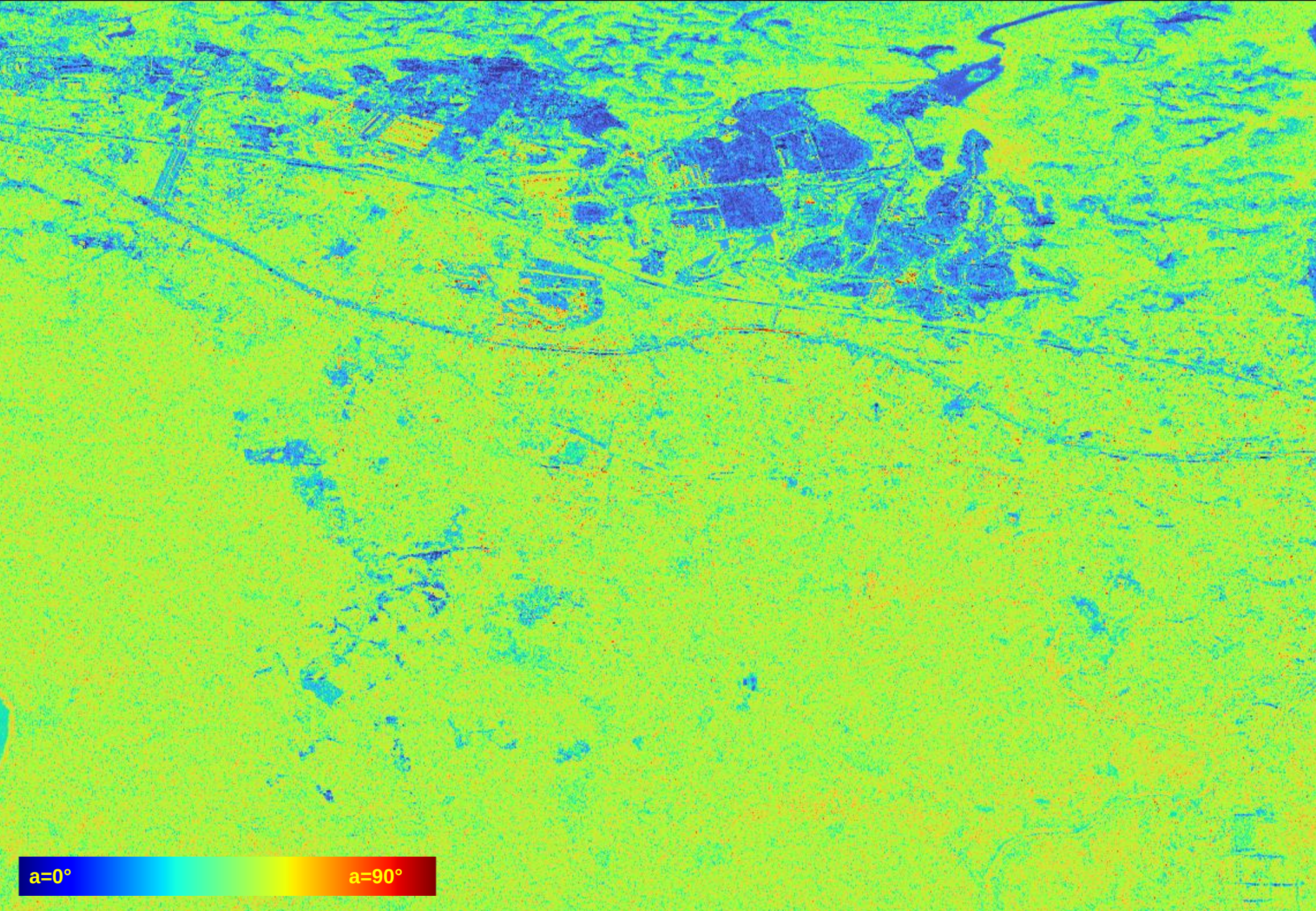
H=0

H=1



A=0

A=1



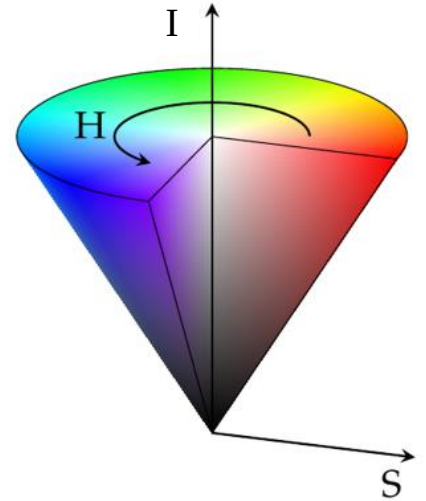
$a=0^\circ$

$a=90^\circ$

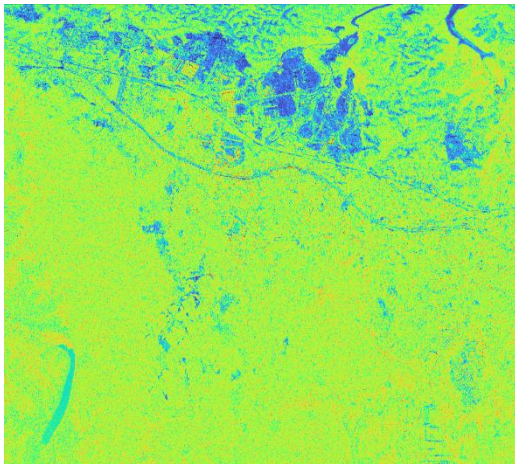
Hue Saturation Intensity (HSI) Color Representation

The HSI color model represents every color with three components: Hue (H), Saturation (S), Intensity (I).

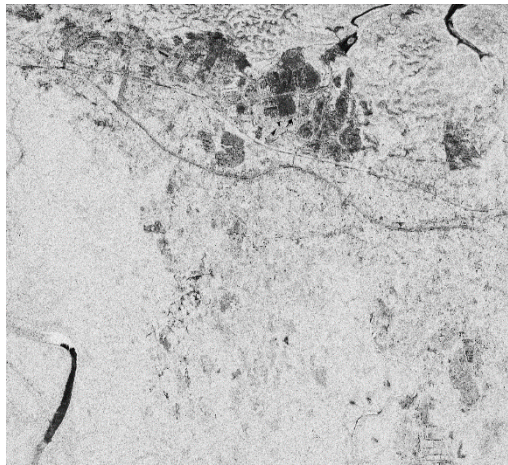
- The Hue component express the color itself in the form of an angle between $[0^\circ, 360^\circ]$. Red is at 0° , green at 120° and blue at 240° .
- The Saturation component express the dominance of the color (e.g. how much the color is diluted with white color). The range of the S is $[0,1]$.
- The Intensity component express how light or dark the color is range is between $[0,1]$. 0 means black, 1 means white.



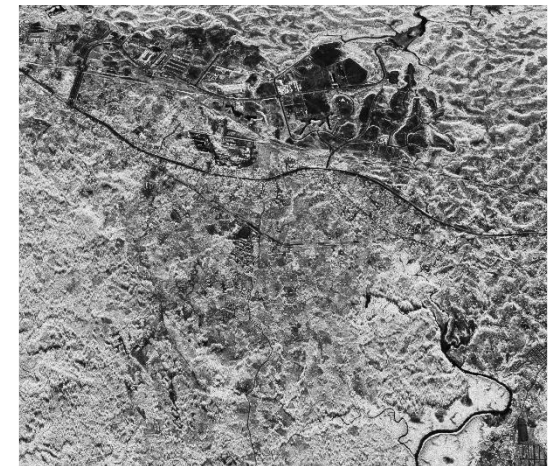
Polarimetric HSI Color Composite: $H=4 \times \text{Mean Alpha angle}$, $S=1-\text{Entropy}$, $I: \text{Total Power scaled from 0 to 1}$



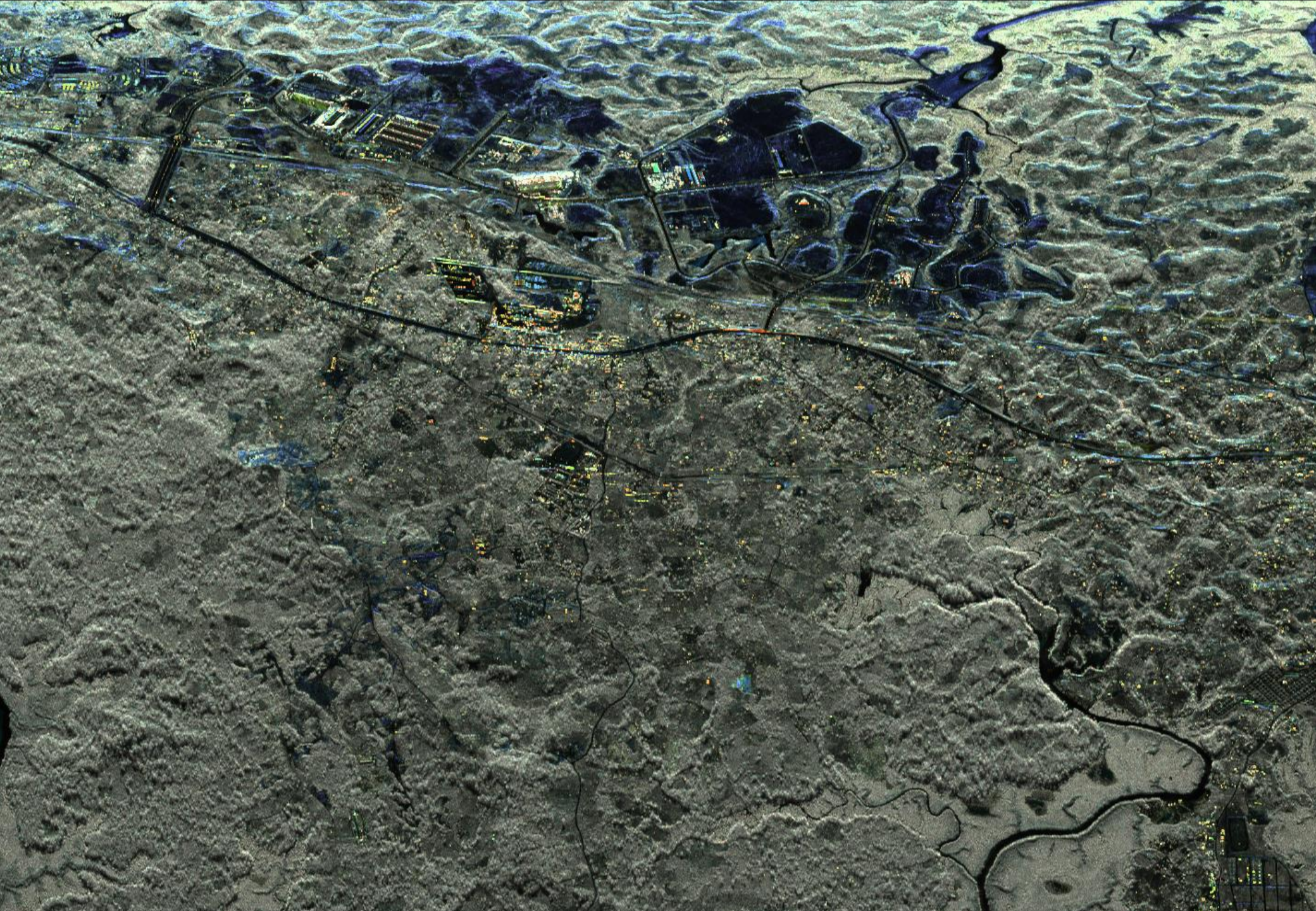
Mean Alpha angle



Entropy



Total Power



The Course

Synthetic Aperture Radar (SAR)

Part 7: SAR Polarimetry 2

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Email: noelia.romerpuig@dlr.de, paloma.saporta@dlr.de, kostas.papathanassiou@dlr.de



ESAMAAP

EEBI  **MASS**